NEAP 2025 U34 EXAM - hi this it

Chemistry (High School - Australia)

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VCE Chemistry Units 3&4

Question and Answer Booklet

2025 Trial Examination

Reading time: 15 minutes

Writing time: 2 hours 30 minutes

Student's Name: _____

Teacher's Name: _____

Approved materials

- One scientific calculator

Materials supplied

- Question and Answer Booklet of 34 pages
- Data Booklet
- Multiple-Choice Answer Sheet

Instructions

- Write **your name** and your **teacher's name** in the spaces above on this page.
- Follow the instructions on your Multiple-Choice Answer Sheet.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this booklet.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

Section A (30 questions, 30 marks) 2–14
Section B (9 questions, 90 marks) 15–34

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Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** or that **best answers** the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this booklet are **not** drawn to scale.

Question 1

The validity of an experiment is

A. improved if numerous measurements are taken.

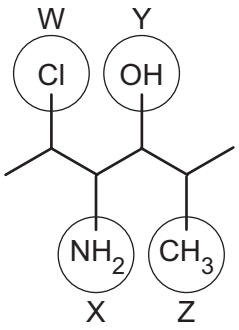
B. assured when the aim of an experiment is achieved using appropriate methods.

C. influenced by a large sample size and repeated measurements.

D. maximised when random and systematic errors are eliminated.

Question 2

Consider the following compound.



Which one of the following would determine the suffix of the compound's IUPAC name?

A. atom W

B. group X

C. group Y

D. group Z

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Question 3

Water is a reactant in

A. the formation of protein from amino acids.

B. the production of starch from glucose.

C. respiration in human cells.

D. photosynthesis in green plants.

Question 4

Glycerol can be produced from the breakdown of

A. triglycerides.

B. polysaccharides.

C. glycogen.

D. proteins.

Question 5

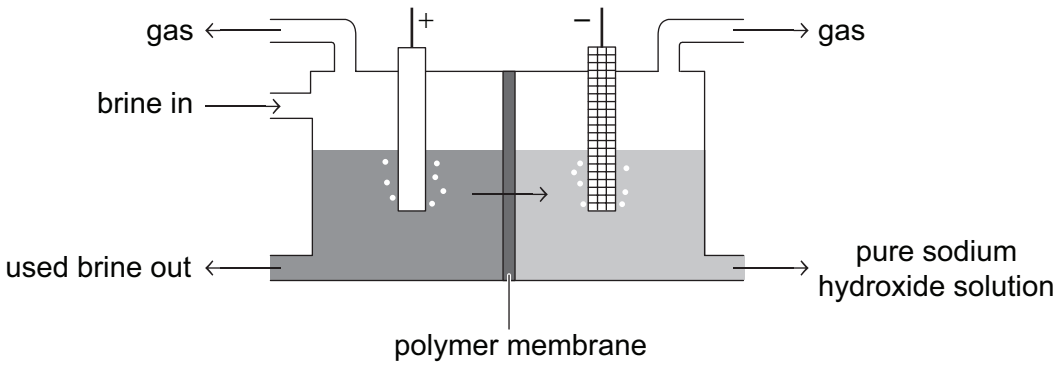
Which one of the following describes features of a secondary cell and a fuel cell?

	Secondary cell	Fuel cell
A.	converts electrical energy to chemical energy	is recharged by an external power supply
B.	always uses inert electrodes	converts chemical energy to electrical energy
C.	usually has porous electrodes that are coated with a catalyst	always has porous, non-metallic electrodes
D.	produces thermal energy and electrical energy	has externally stored reactants fed into it continuously

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Use the following information to answer Questions 6 and 7.

A simplified diagram of an industrial electrolytic membrane cell is shown below. This cell uses a concentrated saltwater electrolyte known as brine and produces a range of important products.



Question 6

Using the electrochemical series, it can be predicted that oxygen gas will be produced at the cell's positive electrode. However, when the cell is operated, chlorine gas is produced at the positive electrode.

Which one of the following statements is correct?

A. Oxygen gas is produced at the negative electrode where hydroxide ions are split apart.

B. Chlorine gas is produced at the positive electrode because chlorine gas is a stronger oxidising agent than oxygen gas.

C. Hydrogen gas is produced at the negative electrode because water is the strongest oxidising agent present.

D. Hydrogen gas is produced at the positive electrode because hydrogen gas is a stronger reducing agent than chloride ions.

Question 7

The polymer membrane is selective, which means that only some substances can pass through it.

Which one of the following species passes through the polymer membrane?

A. chloride ions

B. hydrogen gas

C. hydroxide ions

D. sodium ions

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Question 8

Which one of the following statements is correct for an electrolytic cell but not for a galvanic cell?

A. The cell reaction is a spontaneous redox reaction that generates energy.

B. The products of the cell reaction can be reliably predicted under any conditions.

C. Under no circumstances is the electrode material consumed in the cell reaction.

D. For the cell reaction to move forwards, an input of electrical energy is required.

Question 9

T_A is the temperature at which a substance will burn in air without an external source of ignition; for example, a lit match or a spark.

The magnitude of T_A for a reactant undergoing combustion is closely related to the magnitude of

A. the activation energy of the reaction only.

B. the enthalpy change of the reaction only.

C. both the activation energy and the enthalpy change of the reaction.

D. neither the activation energy nor the enthalpy change of the reaction.

Question 10

Hydrogen gas, methane gas and ethanol can be classified as biofuels when they

A. produce no greenhouse gases when reacted with oxygen gas.

B. are only produced from materials that originated in plants.

C. are derived from recently grown plant or animal material.

D. can no longer be manufactured from fossil fuels.

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Use the following information to answer Questions 11 and 12.

Nitrogen gas reacts with hydrogen gas, according to the following equation.

$$\text{Reaction 1: } N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92.3 \text{ kJ mol}^{-1}$$

At 25°C, the equilibrium constant, K_c, for reaction 1 is 640 M^{-2}.

Consider the reaction represented by the following equation.

$$\text{Reaction 2: } 4NH_3(g) \rightleftharpoons 2N_2(g) + 6H_2(g)$$

Question 11

What is the magnitude of K_c for reaction 2 at 25°C?

A. 2.44×10^{-6}

B. 7.81×10^{-4}

C. 3.13×10^{-3}

D. 3.92×10^{-2}

Question 12

Which one of the following statements is correct?

A. The enthalpy change, ΔH, for reaction 2 is –184.6 kJ mol^{-1}.

B. At 100°C, the value of K_c for reaction 1 is likely to be greater than 640 M^{-2}.

C. In reaction 1, the energy of the products is less than the energy of the reactants.

D. In reaction 2, the concentration of NH_3 is always twice the concentration of N_2.

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Use the following information to answer Questions 13 and 14.

A solution calorimeter containing 200 mL of water was calibrated electrically and produced the following results.

Initial temperature of water	17.5°C
Current applied for 240 seconds	2.5 A
Temperature of water when current ceased	20.1°C
Maximum temperature of water	20.5°C

The calibration factor for the calorimeter is 960 J $°C^{-1}$.

Question 13

What voltage was applied during the calibration?

A. 4.2 V

B. 4.8 V

C. 5.3 V

D. 5.7 V

Question 14

Consider the following statements about the calibration of the calorimeter.

I From the data provided, it is certain that the calorimeter's insulation is very effective.

II The calibration factor would be greater if 250 mL of water, rather than 200 mL, were used in the calorimeter.

III The calorimeter could also be calibrated using aqueous reactants in a chemical reaction of known enthalpy change.

Which of these statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Question 15

Hydrogen gas has been described as a fuel of the future. It has many uses, including being the energy source in fuel cells and being burned in combustion engines.

Which one of the following statements is correct?

A. One advantage of hydrogen fuel is that it does not produce greenhouse gases under standard laboratory conditions (SLC).

B. The energy conversions in a hydrogen fuel cell and a hydrogen-fuelled combustion engine are identical.

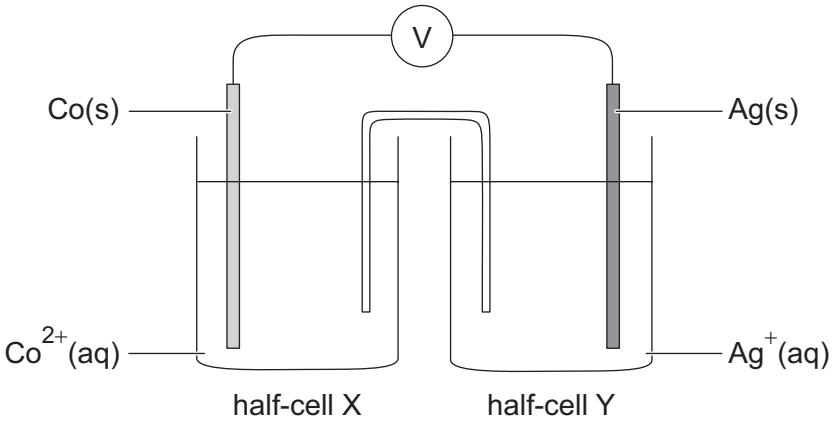
C. For a given mass of hydrogen, a hydrogen fuel cell produces a greater total output of energy than a hydrogen-fuelled combustion engine.

D. The energy content of equal masses of hydrogen gas and octane is roughly the same.

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Use the following information to answer Questions 16 and 17.

The following diagram shows a galvanic cell under SLC.



Question 16

Electrons move from

A. half-cell X to half-cell Y, where reduction occurs.

B. half-cell X to half-cell Y, where oxidation occurs.

C. half-cell Y to half-cell X, where reduction occurs.

D. half-cell Y to half-cell X, where oxidation occurs.

Question 17

Consider the following statements about the cell.

I Positive ions move to half-cell Y along the internal circuit.

II An organic solvent could be used to wet the salt bridge but the solvent must not be able to react with cobalt ions, Co^{2+}, or silver ions, Ag^{+}.

III When the cell reaction is at equilibrium, the cell does not produce electrical energy.

Which of these statements are correct?

A. I and II only

B. I and III only

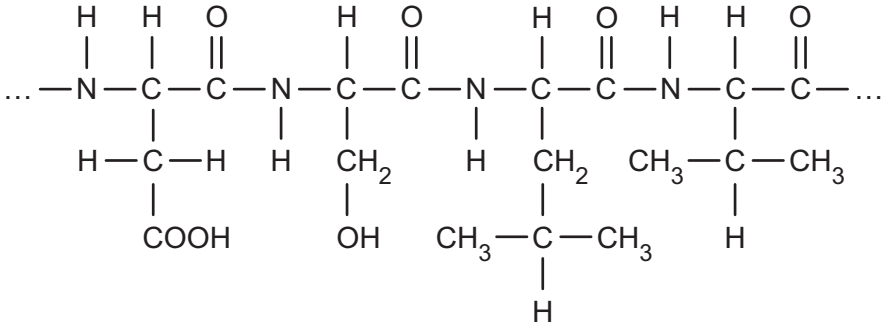
C. II and III only

D. none of I, II or III

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Question 18

The structural formula of a section of an enzyme molecule is shown below.



Which one of the following statements is correct?

A. The section would have been produced from the enzyme by a condensation reaction.

B. The structural formula represents the primary and secondary structure of the section.

C. Heating the section to 60°C would cause the peptide bonds to break.

D. There are four amino acid components and three peptide bonds shown in this section of the enzyme.

Question 19

A component of biodiesel is $C_{11}H_{22}O_2$, and a component of petrodiesel is $C_{12}H_{26}$.

Which one of the following compares features of these compounds?

	$C_{11}H_{22}O_2$	$C_{12}H_{26}$
A.	higher boiling point	higher energy content
B.	lower polarity	higher boiling point
C.	produces more carbon dioxide per mole when burned	lower polarity
D.	lower boiling point	produces more carbon dioxide per mole when burned

Question 20

In a protein, covalent bonding is possible in the

A. primary and secondary structures only.

B. primary and quaternary structures only.

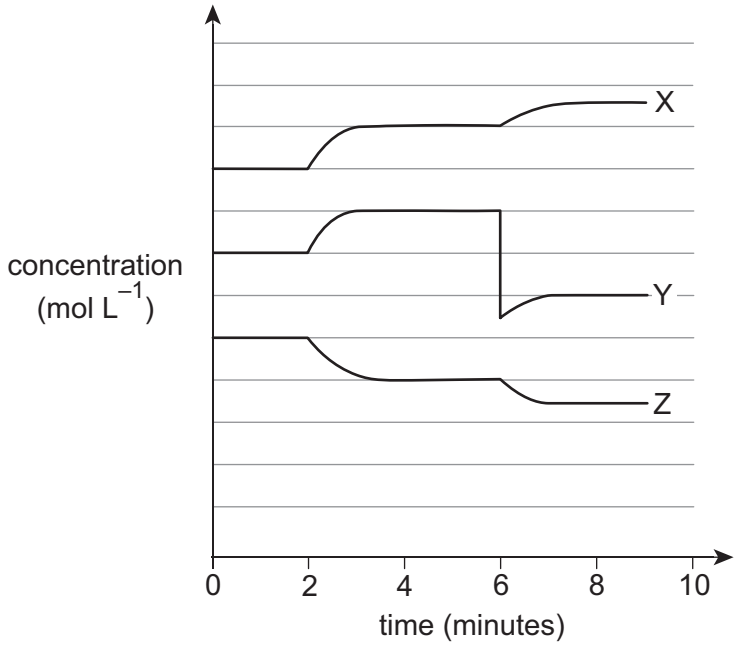
C. primary, tertiary and quaternary structures only.

D. secondary, tertiary and quaternary structures only.

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Use the following information to answer Questions 21 and 22.

The following concentration versus time graph was recorded for an equilibrium reaction in a sealed container. Changes were made to the equilibrium system during the period shown in the graph. The reaction is represented by the equation $X(g) + Y(g) \rightleftharpoons Z(g)$.



Question 21

At two minutes, the equilibrium system was heated.

It can be concluded that the forward reaction is

A. endothermic and more energy is stored in the products than in the reactants.

B. endothermic and more energy is stored in the reactants than in the products.

C. exothermic and more energy is stored in the products than in the reactants.

D. exothermic and more energy is stored in the reactants than in the products.

Question 22

Which one of the following identifies the number of different values of the equilibrium constant, K_c, that could have been obtained in the experiment and the change made to the system at six minutes?

	Number of different values of K_c	Change made at six minutes
A.	2	The volume of the container was decreased.
B.	2	Some of substance Y was removed from the container.
C.	3	The volume of the container was increased.
D.	3	Some of substance Y was removed from the container.

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Use the following information to answer Questions 23 and 24.

The following table shows data about some fuels.

Fuel	Molar mass (g mol^{-1})	Energy released
propane	44	2220 kJ mol^{-1}
butanol	74	36.08 MJ kg^{-1}
ethyne	26	49.9 kJ g^{-1}
pentane	72	49.1 J mg^{-1}

Question 23

When 50 g of the fuel is oxidised completely, which fuel releases the least amount of energy?

A. propane

B. butanol

C. ethyne

D. pentane

Question 24

0.0150 mol of ethyne was oxidised completely to heat a sample of water with an initial temperature of 17.5°C to a final temperature of 65.3°C.

Assuming the complete transfer of energy from the fuel to the water, what was the mass of the water sample?

A. 23.3 g

B. 97.4 g

C. 304 g

D. 383 g

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Question 25

An alcohol compound was analysed using high-performance liquid chromatography (HPLC) with a polar stationary phase and a non-polar solvent.

If the analysis was to be repeated, which one of the following changes would increase the retention time?

A. decreasing the pressure used for the HPLC column

B. using a solvent compound that has lower viscosity

C. using a more loosely packed stationary phase

D. changing the solvent to a compound of high polarity

Question 26

In an experiment, different oxidising and reducing agents were mixed and the results were observed and recorded. Some of the observations are shown in the following table.

Oxidising and reducing agents mixed	Observation
$ClO_2(g)$ and $I^-(aq)$	reaction occurred
$Fe^{3+}(aq)$ and $ClO_2^-(aq)$	no reaction occurred
$Cl_2(g)$ and $ClO_2^-(aq)$	reaction occurred
$I_2(s)$ and $Fe^{2+}(aq)$	no reaction occurred

Based on these observations, which one of the following statements is correct?

A. $ClO_2(g)$ is a stronger oxidising agent than $I_2(s)$ but a weaker oxidising agent than $Fe^{3+}(aq)$.

B. $Cl_2(g)$ is a stronger oxidising agent than $ClO_2(g)$ but a weaker oxidising agent than $I_2(s)$

C. $ClO_2^-(aq)$ is a stronger reducing agent than $Cl^-(aq)$ but a weaker reducing agent than $Fe^{2+}(aq)$.

D. $I^-(aq)$ is a stronger reducing agent than $ClO_2^-(aq)$ but a weaker reducing agent than $Cl^-(aq)$.

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Use the following information to answer Questions 27 and 28.

A galvanic cell was set up with an alkaline electrolyte. The overall cell reaction is represented by the following equation.

$$Zn(s) + 2MnO_2(s) \rightarrow ZnO(s) + Mn_2O_3(s)$$

Question 27

Once the cell is producing electrical energy, the wire and salt bridge are removed, and the contents of both half-cells are put in the same beaker.

Which one of the following best describes what will occur?

A. Electrical energy will be produced because the same spontaneous redox reaction will take place.

B. Electrons will move through the electrolyte because there is no wire present in the beaker.

C. No electrical energy will be produced but the same overall reaction will occur.

D. Both electrical energy and heat energy will be produced.

Question 28

Consider the following statements about the cell.

I The cell is a secondary cell because the products of the energy-producing reaction remain in contact with the electrodes.

II The cell can be recharged by connecting the positive terminal of the power supply to the cell's negative electrode and the negative terminal to the positive electrode.

III In the cell reaction, zinc metal is the reducing agent and undergoes oxidation.

Which of these statements is correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

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Use the following information to answer Questions 29 and 30.

Volumetric analysis of an ethanol solution was performed using the reaction shown in the following equation.

$$2Cr_2O_7^{2-}(aq) + 3C_2H_5OH(aq) + 16H^+(aq) \rightarrow 4Cr^{3+}(aq) + 3CH_3COOH(aq) + 11H_2O(l)$$

A 10.00 mL sample of the ethanol solution was pipetted into a 100.0 mL volumetric flask, and distilled water was added up to the mark. After the flask was shaken, 20.0 mL aliquots were placed into conical flasks and titrated using 0.0495 M $K_2Cr_2O_7$ solution. The average titre was 17.85 mL.

Question 29

Using the data obtained, a student calculated the concentration of the undiluted sample of ethanol solution. Their calculation is shown below.

Step 1: $n(K_2Cr_2O_7) = 0.0495 \times 0.01785$

Step 2: $n(C_2H_5OH) = 1.5 \times 0.0495 \times 0.01785$

Step 3: $c(C_2H_5OH \text{ diluted}) = \dfrac{1.5 \times 0.0495 \times 0.01785}{0.0200}$

Step 4: $c(C_2H_5OH \text{ undiluted}) = \dfrac{1.5 \times 0.0495 \times 0.01785 \times 5}{0.0200}$

Which step of the calculation contains an error?

A. step 1

B. step 2

C. step 3

D. step 4

Question 30

Which one of the following actions would cause the calculated concentration of the ethanol solution to be lower than the actual concentration?

A. rinsing the conical flasks with $K_2Cr_2O_7$ solution prior to use

B. rinsing the conical flasks with water prior to use

C. rinsing the burette with $K_2Cr_2O_7$ solution prior to use

D. rinsing the burette with water prior to use

End of Section A

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Section B

Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Give simplified answers to all numerical questions, with an appropriate number of significant figures; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions; no marks will be given for an incorrect answer unless it is accompanied by details of the working.
- Ensure chemical equations are balanced and that the formulas for individual substances include an indication of state, for example, $H_2(g)$, NaCl(s).
- Unless otherwise indicated, the diagrams in this booklet are **not** drawn to scale.

Question 1 (9 marks)

Hydrogen gas is manufactured industrially according to the reaction shown in the following equation.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \qquad K_c = 5.67\ M^{-2} \text{ at } 1500°C$$

a. Calculate the percentage atom economy for this reaction. 1 mark

b. **i.** Using collision theory, explain why high pressure would achieve the best rate of reaction. 2 marks

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ii. Using Le Chatelier's principle, explain why low pressure would achieve the best yield of hydrogen gas. 2 marks

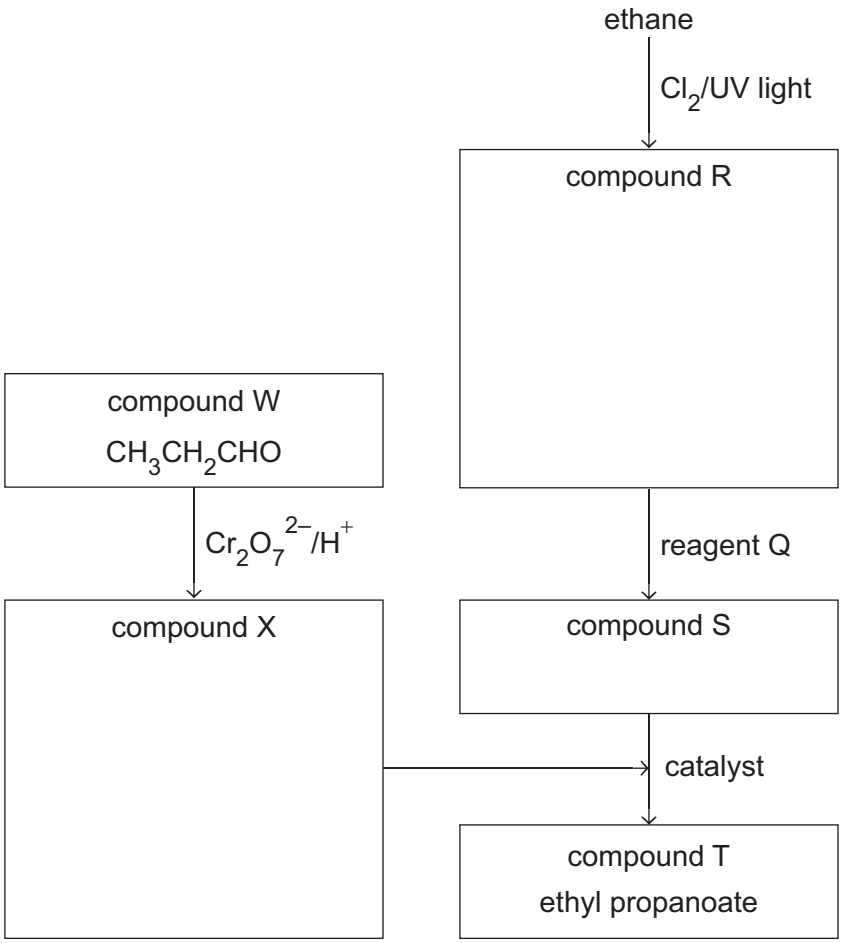
c. A sealed 500.0 mL container at 1500°C contained a gaseous equilibrium mixture that comprised 0.500 mol each of methane, CH_4, water vapour, H_2O, and hydrogen, H_2.

Calculate the amount of carbon monoxide, CO, in the container. 3 marks

d. Describe the circumstances under which hydrogen gas produced by this reaction could be classified as green hydrogen. 1 mark

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Question 2 (9 marks)
The diagram below shows a reaction pathway to produce ethyl propanoate.



a. **i.** Write the IUPAC name of compound W. 1 mark

ii. Identify the type of reaction that occurs when compound W reacts to form compound X. 1 mark

iii. Draw the skeletal formula of compound X in the box provided above. 1 mark

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b. **i.** Identify the type of reaction that occurs when ethane reacts to form compound R. 1 mark

ii. Draw the structural formula of compound R in the box provided on page 17. 1 mark

iii. Identify reagent Q. 1 mark

iv. Write the semi-structural formula of compound S in the box provided on page 17. 1 mark

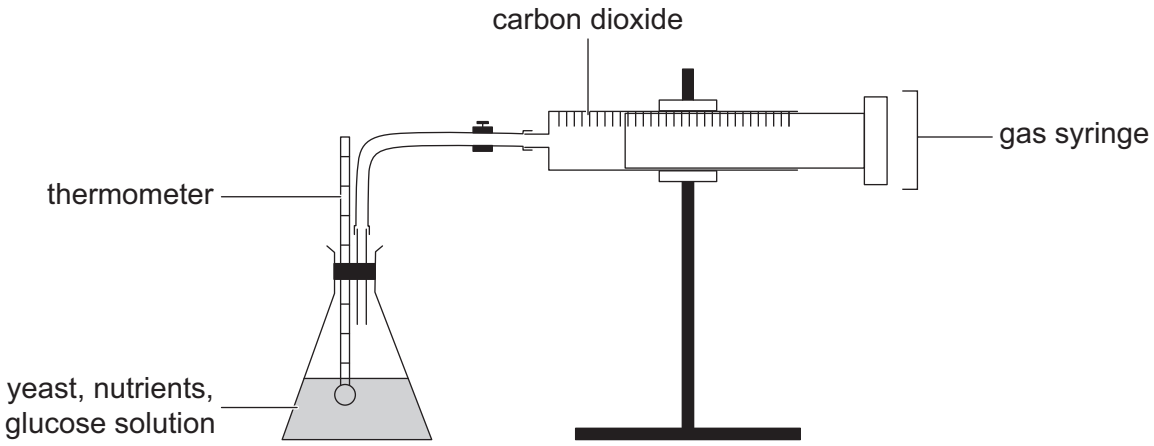
c. Compounds X and S react in the presence of a catalyst.
Write the molecular formula of a suitable catalyst. 1 mark

d. Esters are isomers of carboxylic acids.
State the name of **one** carboxylic acid that is an isomer of ethyl propanoate. 1 mark

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Question 3 (11 marks)

An experiment involving the fermentation of glucose was conducted under standard laboratory conditions (SLC) using the set-up shown in the diagram below.



The results of the experiment are shown in the following table.

Day	1	2	3	4	5	6
Volume of CO_2 gas (mL)	386	534	709	765	793	793

The fermentation reaction that occurred in the flask is represented by the following equation.

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

a. **i.** The molar mass of $C_6H_{12}O_6$ is 180 g mol^{-1}.
Calculate the mass of $C_6H_{12}O_6$ that was consumed in the reaction. 3 marks

ii. Why was it necessary for the experimenter to ensure that the gas in the syringe was at SLC before reading its volume? 1 mark

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b. The experimenter recorded the volume of gas produced until the fermentation reaction stopped. There are two possible reasons why the reaction stopped. One reason is that proteins denature at certain concentrations of ethanol.

i. Identify the evidence that would have enabled the experimenter to deduce that the reaction had stopped. 1 mark

ii. Explain why denatured proteins prevent yeast from fermenting glucose. 2 marks

iii. State the other possible reason why the reaction stopped. 1 mark

iv. Describe another way that the experimenter could have deduced that the reaction had stopped. 1 mark

c. The ethanol produced by this reaction is classified as bioethanol.

Explain why this bioethanol would need to be distilled before it can be used as a transport fuel. 2 marks

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Question 4 (10 marks)

A label on a bottle of cow's milk has the following nutritional information.

Constituent	Quantity per 100 g
protein	3.2 g
fat – total	3.4 g
fat – saturated	2.2 g
carbohydrates – total	5.3 g
carbohydrates – sugars	5.3 g
calcium	110 mg

a. Calculate the energy content of 250 g of cow's milk. 2 marks

b. If left standing, cow's milk will separate into a 'cream' layer, which contains most of the milk's fats, and a 'watery' layer.

Explain why the carbohydrates in cow's milk dissolve in the watery layer. 2 marks

c. One of the components of a protein in cow's milk is the amino acid valine.

Draw the structural formula of valine as a zwitterion. 2 marks

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d. There are special enzyme tablets that can change cow's milk into a dessert. When one of these tablets is dissolved in milk, the milk changes from a liquid to a soft solid after several hours.

Billie dissolved an enzyme tablet in hot milk, believing that using hot milk would cause the soft solid to form faster. However, the hot milk did not change from a liquid to a soft solid.

i. Explain the chemical reasoning that Billie used to conclude that hot milk would make the reaction happen faster. 2 marks

ii. Explain why the hot milk did not change from a liquid to a soft solid. 2 marks

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Question 5 (10 marks)

Research is being undertaken to develop methods of using the greenhouse gas carbon dioxide to produce useful compounds instead of releasing the gas into the atmosphere.

a. Diesel fuel is used in some electrical generators that release carbon dioxide.

i. Write the balanced equation for the complete combustion of $C_{18}H_{38}$, which is a typical component of diesel fuel. 2 marks

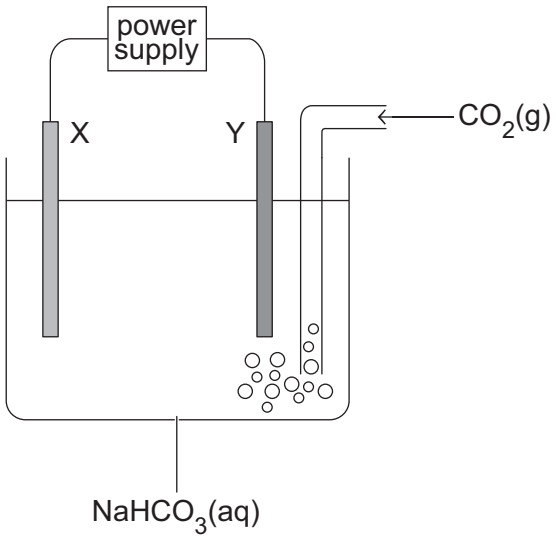
ii. The heat of combustion of a particular diesel fuel is 45.0 kJ g^{-1}. When 75.0 g of this diesel fuel is burnt in a generator, it produces enough electrical energy to run a device at 2.5 V and 9.7 A for 17.2 hours.

Calculate the efficiency of the generator. 3 marks

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b. The diagram below shows an electrolytic cell that converts carbon dioxide gas into ethanol under alkaline conditions using the following chemical reaction.

$$2CO_2(g) + 3H_2O(l) \rightarrow C_2H_5OH(l) + 3O_2(g)$$



i. Write the half-equation for the reaction that occurs at electrode Y. 2 marks

ii. In the table below, draw a tick in **one** box to indicate the reaction type that occurs at electrode Y and the polarity of electrode Y. 1 mark

	Oxidation	**Reduction**
Positive		
Negative		

iii. Outline **one** observation that could be made at electrode X as the electrolysis proceeds. 1 mark

iv. Electrode X is made of graphite, and electrode Y is made of a copper–zinc alloy.

Why is electrode Y made of a copper–zinc alloy rather than a less expensive material such as graphite? 1 mark

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Question 6 (14 marks)

a. The melting points of different unsaturated fatty acids are shown in the table below.

Fatty acid	Melting point (°C)
palmitoleic acid	1.0
oleic acid	16.0
linoleic acid	–5.0
linolenic acid	–11.0

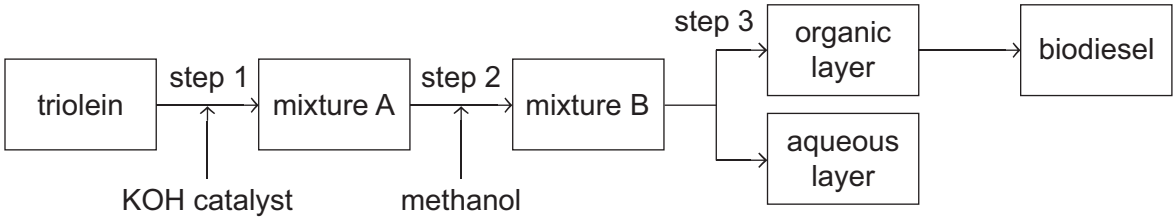
i. With reference to structure and bonding, explain why oleic acid has a higher melting point than palmitoleic acid. 2 marks

ii. The molar mass of linolenic acid is 278 g mol^{-1}.

Determine the mass of iodine, I_2, that would react with 10.00 g of linolenic acid. 3 marks

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b. Oleic acid is used in the formation of triolein, which is a triglyceride found in vegetable oils used for cooking. Rather than being discarded as waste, used vegetable oil can be converted into biodiesel. The following flow chart shows a process that produces biodiesel from used vegetable oil via transesterification.



i. In step 1, potassium hydroxide, KOH, is used as a catalyst to break down the triolein.

Name **one** substance that would be found in mixture A other than oleic acid, KOH and water. 1 mark

ii. In step 2, methanol reacts with oleic acid to produce methyl oleate, which is a biodiesel.

Write the balanced equation for this reaction. States are **not** required. 1 mark

iii. Mixture B contains substances that are soluble in water. It also contains biodiesel, which is not soluble in water. Step 3 separates this mixture.

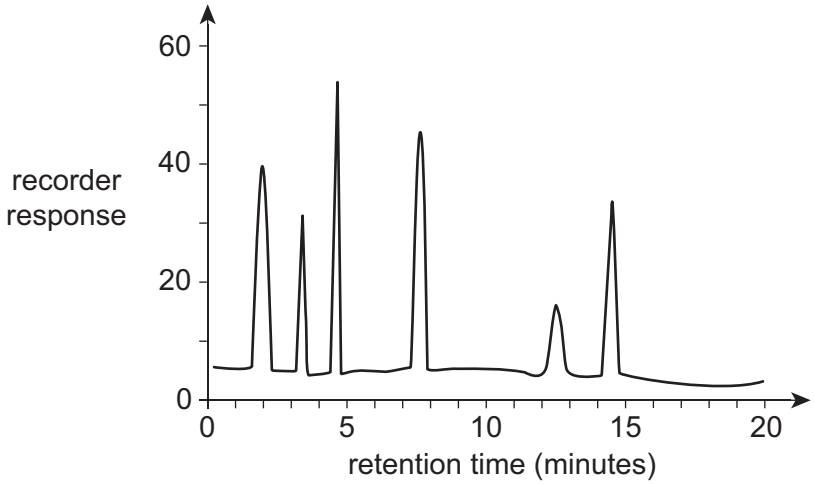
Name **one** non-organic compound that is present in the aqueous layer after step 3. 1 mark

iv. This process demonstrates two green chemistry principles: catalysis and the use of renewable feedstocks. The enzyme lipase can be used as a substitute catalyst in step 1.

Outline how using lipase would enable the process to conform more closely to green chemistry principles. 2 marks

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c. A mixture of fatty acids was analysed using high-performance liquid chromatography (HPLC). The following output was produced.



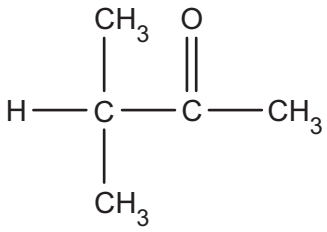
i. Explain how the fatty acids in the mixture can be identified using these results. 2 marks

ii. Explain how the concentration of the fatty acid with a retention time of 7.8 minutes can be determined. 2 marks

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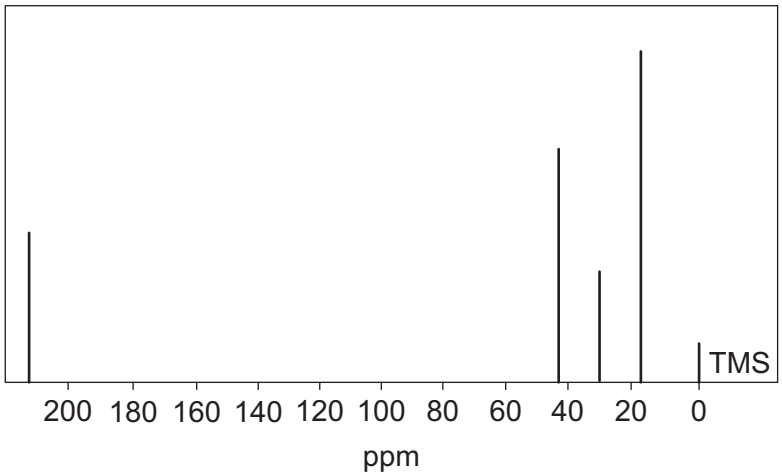
Question 7 (9 marks)

Instrumental analysis of a particular organic compound was performed. The structural formula of the compound is shown below.



a. State the IUPAC name of this compound. 1 mark

b. The carbon-13 nuclear magnetic resonance (^{13}C NMR) spectrum of the compound is shown below.

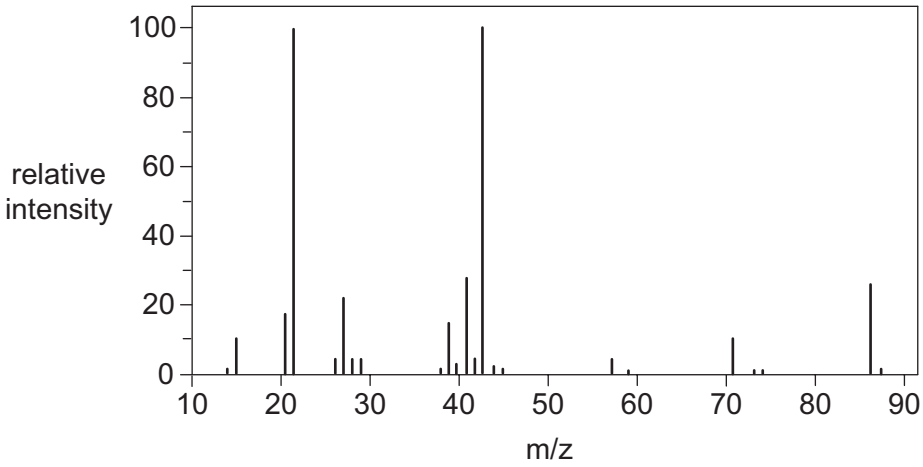


i. Why does the spectrum show only four peaks when the compound contains five carbon atoms? 1 mark

ii. Which group of atoms is responsible for the peak at 18 ppm? 1 mark

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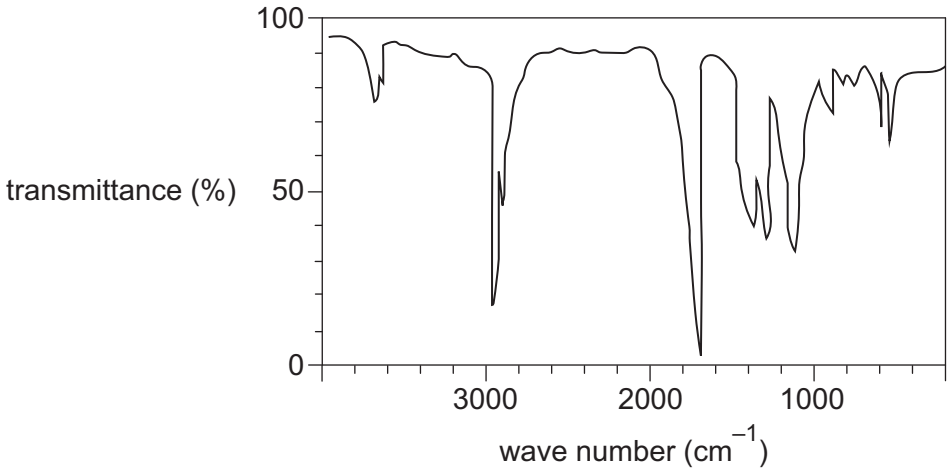
c. The mass spectrum of the compound is shown below.



i. What is the m/z value of the molecular ion? 1 mark

ii. State the formula of the fragment with a m/z value of 43. 1 mark

d. The infrared (IR) spectrum of the compound is shown below.



i. How is the signal at 1700 cm^{-1} consistent with the structure of the compound? 1 mark

ii. Carboxylic acids have a similar functional group to the compound.
What evidence from the IR spectrum indicates that the compound is not a carboxylic acid? 1 mark

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e. The following table shows data from the proton nuclear magnetic resonance (^{1}H NMR) spectrum of the compound.

Component	Chemical shift (ppm)	Relative peak area	Peak splitting
A	1.1	6	doublet
B	2.0	3	singlet
C	2.6	1	?

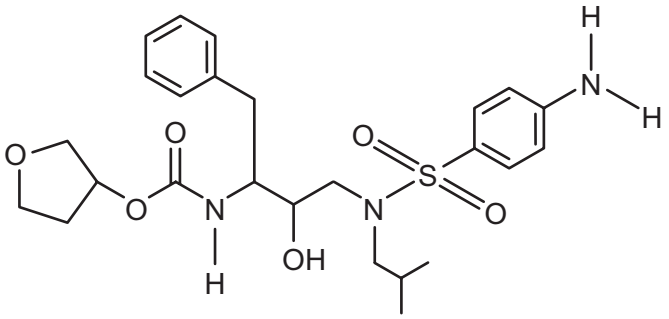
i. Which component (A, B or C) is consistent with a methyl group with one neighbouring hydrogen? 1 mark

ii. What is the expected peak splitting for component C? 1 mark

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Question 8 (10 marks)

Acquired immunodeficiency syndrome (AIDS) is caused by the human immunodeficiency virus (HIV). The virus infects human cells and uses the enzyme HIV protease to make the cells produce more virus particles. Researchers have developed competitive enzyme inhibitors that block the activity of HIV protease and prevent more virus particles from forming. The structural formula of one of these inhibitor molecules is shown below.



a. **i.** On the structural formula above, draw a circle around **one** functional group and write the name of the functional group next to the circle. 1 mark

ii. On the structural formula above, draw an arrow pointing to **one** carbon atom that is **not** chiral. In the space below, state why the selected carbon atom is not chiral. 2 marks

b. **i.** One model that explains enzyme action is the lock-and-key mechanism.
Describe this model. 3 marks

ii. Explain how competitive enzyme inhibitors block the function of an enzyme. 2 marks

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c. Researchers who are developing competitive enzyme inhibitors against HIV protease could display their research as a scientific poster.

Explain the purpose of a scientific poster. 2 marks

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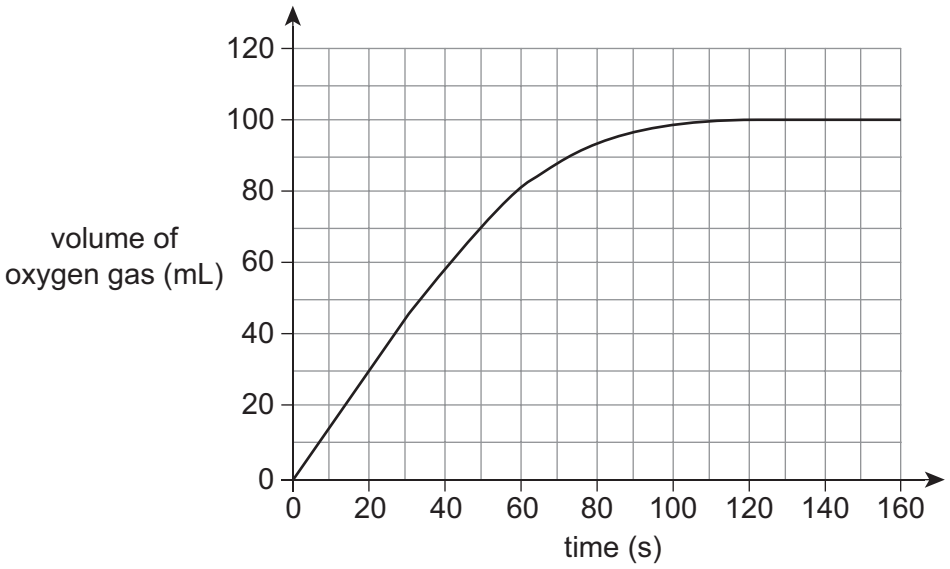
Question 9 (8 marks)
Hydrogen peroxide solution decomposes according to the following equation.

$$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$$

a. According to the electrochemical series, this reaction should occur because H_2O_2 is both an oxidising agent and a reducing agent.
Write the half-equation for the oxidation reaction that occurs during the decomposition of hydrogen peroxide. 1 mark

Hydrogen peroxide decomposes rapidly if finely divided or powder-like platinum metal is added as a catalyst.

b. In an experiment, 50.0 mL of 0.161 M hydrogen peroxide solution was placed in a flask with 1.0 g of finely divided platinum. The volume of oxygen gas produced was measured at 10 second intervals. The results are shown in the following graph.



i. Why did the platinum need to be finely divided? 1 mark

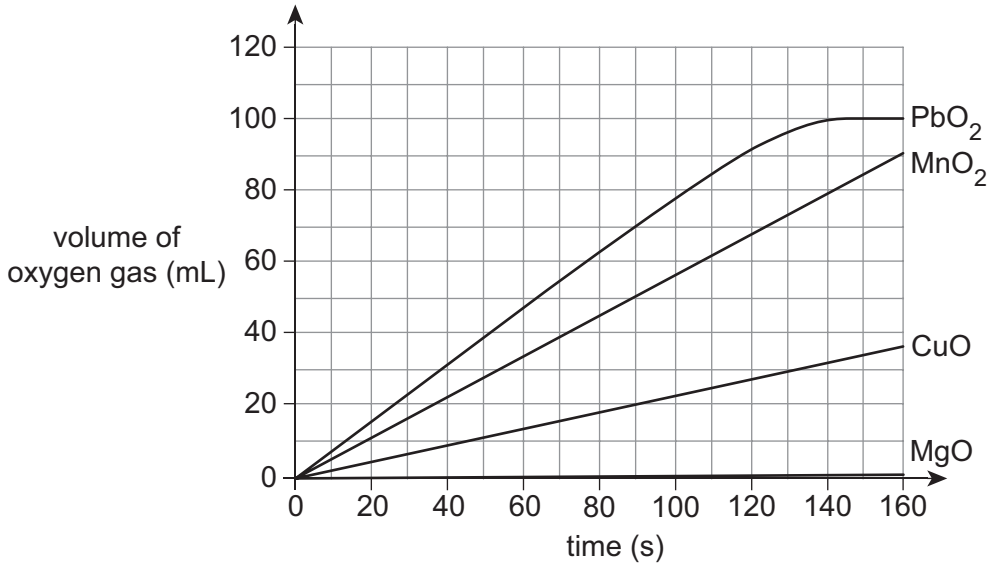
ii. With reference to the rate of reaction, describe what is happening in the 0–50 s interval. 1 mark

iii. With reference to the rate of reaction, describe what is happening in the 110–160 s interval. 1 mark

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c. Platinum is very expensive. To find a substitute for the platinum catalyst, a second experiment was performed. The original experiment was repeated under identical conditions except that, in separate trials, the platinum was replaced with 1.0 g of one of the following compounds: copper(II) oxide, lead(IV) oxide, magnesium oxide and manganese(IV) oxide.

The results of each trial are shown in the following graph.



i. Write an aim for the second experiment. 1 mark

ii. Identify the dependent variable in the second experiment. 1 mark

iii. Write a conclusion for the second experiment. 2 marks

End of examination questions

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